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QUANTILES, PERFORM LOCAL-REGRESSION AND CONSTRUCT
KD-TREES IN PARALLEL.***

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1. Transformations with Chebyshev's Polynomials. In mathematics the Chebyshev's polynomials, named after Pafnuty Chebyshev, are a sequence of orthogonal polynomials which can be defined recursively. The Chebyshev's polynomials of the first kind are defined by the recurrence relation $T_0(t) = 1$; $T_1(t) = t$; and $T_{j+1}(t) = 2t \cdot T_j(t) - T_{j-1}(t)$ for $j > 0$ and $t \in \mathbb{R}$. The expressions of first few non-trivial Chebyshev's polynomials for $t \in \mathbb{R}$ are as follows,

$$\begin{aligned} T_2(t) &= 2t^2 - 1 \\ T_3(t) &= 4t^3 - 3t \\ T_4(t) &= 8t^4 - 8t^2 + 1 \\ T_5(t) &= 16t^5 - 20t^3 + 5t \end{aligned}$$

The expression of a general j^{th} order Chebyshev's polynomial can be obtained recursively as,

$$T_j(t) = \frac{j}{2} \sum_{j'=0}^{\lfloor \frac{j}{2} \rfloor} (-1)^{j'} \frac{(j-j'-1)!}{j'!(j-j')!} (2t)^{j-2j'}, \text{ for } j \in \mathbb{N}^+ \text{ and } t \in \mathbb{R}.$$

From this expression, we realize an odd-degree Chebyshev's polynomial $T_{2j-1}(t)$ is a non-zero linear combination of $t, t^3, \dots, t^{2j-1}$ for $t \in \mathbb{R}$, and an even-degree Chebyshev's polynomial $T_{2j}(t)$ is a non-zero linear combination of $1, t^2, t^4, \dots, t^{2j}$ for $t \in \mathbb{R}$. Now for $j \in \mathbb{N}^+$ and $z \in (0, 1)$, we have the following trigonometric identities:

$$\begin{aligned} T_{2j-1}(\cos(z)) &= \cos((2j-1)z) \\ T_{2j-1}(\sin(z)) &= (-1)^{j-1} \sin((2j-1)z). \end{aligned}$$

1.1. 1-dimensional transformations with Chebyshev's Polynomials. Let $c_{2j-1}(z) = \cos((2j-1)z)$ and $c_{2j}(z) = \sin((2j-1)z)$ for $j \in \mathbb{N}^+$, also let, $c^{2j-1}(z) = \cos^{2j-1}(z)$ and $c^{2j}(z) = \sin(z) \cos^{2j-2}(z)$ for $j \in \mathbb{N}^+$. Observe that, $c_1(z) = c^1(z) = \cos(z)$ and $c_2(z) = c^2(z) = \sin(z)$. For $J \in \mathbb{N}^+$, let us define vector valued functions:

$$\begin{aligned} \mathbf{c}_{(1:2J)}(z) &= (c_1(z), c_2(z), \dots, c_{2J}(z)), \text{ and} \\ \mathbf{c}^{(1:2J)}(z) &= (c^1(z), c^2(z), \dots, c^{2J}(z)). \end{aligned}$$

Also, for consistency in future results, for $J \in \mathbb{N}^+$ we define $\tilde{\mathbf{c}}_{(1:2J)}(z, J) = \mathbf{c}_{(1:2J)}(z)$ and $\tilde{\mathbf{c}}^{(1:2J)}(z, J) = \mathbf{c}^{(1:2J)}(z)$. Now, we have the following lemma:

LEMMA 1.1. *Given $J \in \mathbb{N}^+$, $\exists$ linear transformation $\tau_J^{(A)} : \mathbb{R}^{2J} \rightarrow \mathbb{R}^{2J}$, such that:*

$$\mathbf{c}_{(1:2J)}(z) = \tau_J^{(A)}(\mathbf{c}^{(1:2J)}(z)).$$

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Proof. First, remember we have the following Chebyshev's identities for Cosines:

$$\mathbf{c}_{2j-1}(z) = \cos((2j-1)z) = T_{2j-1}(\cos(z)).$$

Here $T_{2j-1}(t)$ is the $(2j-1)^{th}$ degree Chebyshev's polynomial, and we have the following expression of $T_{2j-1}(t)$ for $j \in \mathbb{N}^+$:

$$T_{2j-1}(t) = \frac{(2j-1)}{2} \sum_{j'=1}^j (-1)^{j-j'} \frac{(j+j'-2)!}{(j-j')!(2j'-1)!} (2t)^{2j'-1}$$

If we replace t with $\cos(z)$, we get for $j \in \mathbb{N}^+$:

$$\begin{aligned} \mathbf{c}_{2j-1}(z) &= \cos((2j-1) \cdot z) = T_{2j-1}(\cos(z)) \\ &= \sum_{j'=1}^j (-1)^{j-j'} \frac{2^{2j'-2}(2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \cos^{2j'-1}(z) \\ &= \sum_{j'=1}^j (-1)^{j-j'} \frac{2^{2j'-2}(2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \cdot \mathbf{c}_{2j'-1}(z). \end{aligned}$$

For $j \in \{1, 2, \dots, J\}$, define:

$$\begin{aligned} a_{2j-1, 2j'-1} &= (-1)^{j-j'} \frac{2^{2j'-2}(2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \text{ for } 1 \leq j' \leq j, \\ a_{2j-1, 2j'-1} &= 0 \text{ for } j < j' \leq J \text{ and} \\ a_{2j-1, 2j'} &= 0 \text{ for } 1 \leq j' \leq J. \end{aligned}$$

Then, from the expression of $\mathbf{c}_{2j-1}(z)$, we can say that:

$$\mathbf{c}_{2j-1}(z) = \sum_{j'=1}^{2J} a_{2j-1, j'} \cdot \mathbf{c}^{j'}(z) \text{ for } 1 \leq j \leq J.$$

Now remember, we also have the Chebyshev's identities for Sines:

$$\mathbf{s}_{2j}(z) = \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z))$$

In the previous expression of $T_{2j-1}(z)$, if we replace x with $\sin(z)$, we get for $j \in \mathbb{N}^+$:

$$\begin{aligned} \mathbf{s}_{2j}(z) &= \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z)) \\ &= \sum_{j'=1}^j (-1)^{j'-1} \frac{2^{2j'-2}(2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin^{2j'-1}(z). \end{aligned}$$

Now, observe that $\sin^{2j'-1}(z) = \sin(z) - \sum_{j''=1}^{j'-1} \sin(z) \cos^{2j''}(z)$ for $j' > 1$, then for

69 $j > 1$, we have:

$$\begin{aligned}
70 \quad & \mathbf{c}_{2j}(z) = \sin((2j-1) \cdot z) = (-1)^{j-1} T_{2j-1}(\sin(z)) \\
71 \quad & = (2j-1) \sin(z) + \sum_{j'=2}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin^{2j'-1}(z) \\
72 \quad & = (2j-1) \sin(z) \\
73 \quad & + \sum_{j'=2}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \left(\sin(z) - \sum_{j''=1}^{j'-1} \sin(z) \cos^{2j''}(z) \right) \\
74 \quad & = \left(\sum_{j'=1}^j (-1)^{j'-1} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \right) \sin(z) \\
75 \quad & + \sum_{j'=2}^j \sum_{j''=1}^{j'-1} \left((-1)^{j'} \frac{2^{2j'-2} (2j-1)(j+j'-2)!}{(j-j')!(2j'-1)!} \sin(z) \cos^{2j''}(z) \right) \\
76 \quad & = \left(\sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \sin(z) \\
77 \quad & + \sum_{j'=2}^j \left(\sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \sin(z) \cos^{2j'-2}(z) \\
78 \quad & = \left(\sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \cdot \mathbf{c}^2(z) \\
79 \quad & + \sum_{j'=2}^j \left(\sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \right) \cdot \mathbf{c}^{2j'}(z). \\
80 \quad &
\end{aligned}$$

81 Given $j \in \{1, 2, \dots, J\}$, we define:

$$\begin{aligned}
82 \quad & a_{2j, 2j'-1} = 0 \text{ for } 1 \leq j' \leq J, \\
83 \quad & a_{2j, 2} = \sum_{j''=1}^j (-1)^{j''-1} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \text{ and} \\
84 \quad & a_{2j, 2j'} = \sum_{j''=j'}^j (-1)^{j''} \frac{2^{2j''-2} (2j-1)(j+j''-2)!}{(j-j'')!(2j''-1)!} \text{ for } 2 \leq j' \leq j, \\
85 \quad & a_{2j, 2j'} = 0 \text{ for } j+1 \leq j' \leq J.
\end{aligned}$$

87 Then, from the expression of $\mathbf{c}_{2j}(z)$, we can also say that:

$$88 \quad \mathbf{c}_{2j}(z) = \sum_{j'=1}^{2J} a_{2j, j'} \cdot \mathbf{c}^{j'}(z) \text{ for } 1 \leq j \leq J.$$

90 Hence, in general we can say:

$$91 \quad \mathbf{c}_j(z) = \sum_{j'=1}^{2J} a_{j, j'} \cdot \mathbf{c}^{j'}(z) \text{ for } 1 \leq j \leq 2J.$$

93 Define the $2J \times 2J$ matrix $\mathcal{A}_J^{(A)} = \{\{a_{j,j'}\}\}_{1 \leq j \leq 2J, 1 \leq j' \leq 2J}$. Then, we can say
 94 that:

$$95 \quad \mathbf{c}_{(1:2J)}(z) = \mathcal{A}_J^{(A)} \cdot \mathbf{c}^{(1:2J)}(z).$$

97 So, we proved existence of a linear transformation $\tau_J^{(A)} : \mathbb{R}^{2J} \rightarrow \mathbb{R}^{2J}$, such that:

$$98 \quad \mathbf{c}_{(1:2J)}(z) = \tau_J^{(A)}(\mathbf{c}^{(1:2J)}(z)).$$

100 with the explicit representation $\tau_J^{(A)}(\mathbf{v}) = \mathcal{A}_J^{(A)} \cdot \mathbf{v}$. □

101 *Remark:* Form the alternate definition we have:

$$102 \quad \tilde{\mathbf{c}}_{(1:2J)}(z, J) = \tau_J^{(A)}(\tilde{\mathbf{c}}^{(1:2J)}(z, J)).$$

104 **1.2. 2-dimensional transformations with Chebyshev's Polynomials.** Let
 105 us introduce another set of functions for $j \in \mathbb{N}^+$, given $x \in \mathbb{R}$ and $J' \in \mathbb{N}^+$: a fixed
 106 positive integer.

$$107 \quad \dot{c}_j(z, J') = \cos(2 \cdot (j-1) \cdot J' \cdot z) = \cos((j-1) \cdot (2J'z)),$$

$$108 \quad \dot{c}^j(z, J') = \cos^{j-1}(2J'z).$$

110 Also, let us define vector valued functions for $J \in \mathbb{N}^+$:

$$111 \quad \dot{\mathbf{c}}_{(1:J)}(z, J') = (\dot{c}_1(z, J'), \dot{c}_2(z, J'), \dots, \dot{c}_J(z, J')),$$

$$112 \quad \dot{\mathbf{c}}^{(1:J)}(z, J') = (\dot{c}^1(z, J'), \dot{c}^2(z, J'), \dots, \dot{c}^J(z, J')).$$

114 Then, we have the following lemma:

115 **LEMMA 1.2.** *Given a fixed $J' \in \mathbb{N}^+$, for $J \in \mathbb{N}^+$, there exist linear transformation*
 116 $\tau_J^{(B)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$, *such that:*

$$117 \quad \dot{\mathbf{c}}_{(1:J)}(z, J') = \tau_J^{(B)}(\dot{\mathbf{c}}^{(1:J)}(z, J')).$$

119 *Proof.* First given any $J' \in \mathbb{N}^+$, observe $\dot{c}_0(z, J') = 1$, $\dot{c}_1(z, J') = \cos(2J'z)$ and
 120 $\dot{c}^0(z, J') = 1$, $\dot{c}^1(z, J') = \cos(2J'z)$ for any $x \in \mathbb{R}$. For $j > 2$, we again use the
 121 Chebyshev's identities. For a fixed $J' \in \mathbb{N}^+$, we have for $x \in \mathbb{R}$:

$$122 \quad \dot{c}_j(z, J') = \cos((j-1) \cdot (2J'z)) = T_{j-1}(\cos(2J'z)).$$

124 Here $T_j(z)$ is the j^{th} degree Chebyshev's polynomial, and remember we have the
 125 following expression of $T_j(z)$ for $j > 1$,

$$126 \quad T_j(t) = \frac{j}{2} \sum_{k=0}^{\lceil \frac{j}{2} \rceil} (-1)^k \frac{(j-k-1)!}{k!(j-k)!} (2t)^{j-2k}.$$

128 If we replace t with $\cos(2J'z)$ and j with $j-1$, we get for $j \in \mathbb{N}^+$:

$$129 \quad \dot{c}_j(z, J') = \cos((j-1) \cdot (2J'z)) = T_{j-1}(\cos(2J'z))$$

$$130 \quad = \sum_{k=0}^{\lceil \frac{j-1}{2} \rceil} (-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \cos^{j-2k-1}(2J'z)$$

$$131 \quad = \sum_{k=0}^{\lceil \frac{j-1}{2} \rceil} \left((-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \right) \cdot \dot{c}^{j-2k}(z, J').$$

133 Let us define:

$$\begin{aligned}
 134 \quad & a_{1,1} = 1, \quad a_{1,j} = 0 \text{ for } 2 \leq j \leq J; \\
 135 \quad & a_{2,1} = 0, \quad a_{2,2} = 1, \quad a_{2,j} = 0 \text{ for } 2 < j \leq J; \\
 136 \quad & a_{j,k} = (-1)^k \frac{2^{j-2k-2} j(j-k-2)!}{k!(j-k-1)!} \text{ if } j \geq 2, \quad k \leq j, \quad \frac{j-k-1}{2} \in \mathbb{N}^+; \\
 137 \quad & a_{j,k} = 0 \text{ if } j \geq 2, \quad k \leq j, \quad \frac{j-k-1}{2} \notin \mathbb{N}^+ \text{ or } j < k \leq J. \\
 138
 \end{aligned}$$

139 Then, from the expression of $\dot{\mathbf{c}}_j(z, J')$, we can say that:

$$140 \quad \dot{\mathbf{c}}_j(z, J') = \sum_{k=1}^J a_{j,k} \cdot \dot{\mathbf{c}}^j(z, J') \text{ for } 1 \leq j \leq J.$$

142 Define $J \times J$ matrix $\mathcal{A}_J^{(B)} = \{\{a_{j,k}\}\}_{1 \leq j \leq J, 1 \leq k \leq J}$. Then, we can say:

$$143 \quad \dot{\mathbf{c}}_J(z, J') = \mathcal{A}_J^{(B)} \cdot \dot{\mathbf{c}}^J(z, J').$$

145 So, we proved existence of a linear transformation $\tau_J^{(B)} : \mathbb{R}^J \rightarrow \mathbb{R}^J$, such that:

$$146 \quad \dot{\mathbf{c}}_J(z, J') = \tau_J^{(B)}(\dot{\mathbf{c}}^J(z, J')).$$

148 with the explicit representation $\tau_J^{(B)}(\mathbf{x}) = \mathcal{A}_J^{(B)} \cdot \mathbf{x}$. □

149 **1.3. Transformations on matrices.** Consider two integers $J_1 \in \mathbb{N}^+$ and $J_2 \in$
150 $\mathbb{N}^+$. Define:

$$151 \quad \mathbb{N}_{\{J_1, J_2\}} := \{1, \dots, 2J_1\} \times \{1, \dots, J_2\}.$$

153 Define 2 dimensional sequential functions:

$$\begin{aligned}
 154 \quad & \tilde{\mathbf{c}}_{j_1, j_2}(z, J_1, J_2) = \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1). \\
 155 \quad & \tilde{\mathbf{c}}^{j_1, j_2}(z, J_1, J_2) = \mathbf{c}^{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1).
 \end{aligned}$$

157 Also, define matrix-valued functions:

$$\begin{aligned}
 158 \quad & \tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \{\{\tilde{\mathbf{c}}_{j_1, j_2}(z, J_1, J_2)\}\}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}. \\
 159 \quad & \tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \{\{\tilde{\mathbf{c}}^{j_1, j_2}(z, J_1, J_2)\}\}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}. \\
 160
 \end{aligned}$$

161

162 **LEMMA 1.3.** *Given integers $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$, there exist linear transfor-*
163 *mation $\tau_{J_1, J_2}^{A, B} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2}$, such that:*

$$164 \quad \tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \tau_{J_1, J_2}^{A, B}(\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)).$$

166 *Proof.* Let us consider the j_2^{th} column of the matrix $\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)$
167 which is the array:

$$\begin{aligned}
 168 \quad & \tilde{\mathbf{c}}^{(1:2J_1), j_2}(z, J_1, J_2) = (\tilde{\mathbf{c}}^{1, j_2}(z, J_1, J_2), \dots, \tilde{\mathbf{c}}^{2J_1, j_2}(z, J_1, J_2)) \\
 169 \quad & = (\mathbf{c}^1(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1), \dots, \mathbf{c}^{2J_1}(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1)) \\
 170 \quad & = (\mathbf{c}^1(z), \dots, \mathbf{c}^{2J_1}(z)) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1) = \mathbf{c}^{(1:2J_1)}(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1).
 \end{aligned}$$

So, $\tilde{\mathbf{c}}^{(1:2J_1), j_2}(z, J_1, J_2)$ is a scalar multiple of the array $\mathbf{c}^{(1:2J_1)}(z)$ with multiplier $\dot{\mathbf{c}}^{j_2}(z, J_1)$. Then, for the transformation $\tau_J^{(A)}$ in [Lemma 1.1](#) (with $J = J_1$), we have:

$$\begin{aligned} \tau_{J_1}^{(A)}(\tilde{\mathbf{c}}^{(1:2J_1), j_2}(z, J_1, J_2)) &= \tau_{J_1}^{(A)}(\mathbf{c}^{(1:2J_1)}(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1)) \\ &= \tau_{J_1}^{(A)}(\mathbf{c}^{(1:2J_1)}(z)) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1) = \mathbf{c}_{(1:2J_1)}(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1) \\ &= \tilde{\mathbf{c}}_{(1:2J_1), 0}^{0, j_2}(z, J_1, J_2) = (\tilde{\mathbf{c}}_{1, 0}^{0, j_2}(z, J_1, J_2), \dots, \tilde{\mathbf{c}}_{2J_1, 0}^{0, j_2}(z, J_1, J_2)); \\ \text{when: } \tilde{\mathbf{c}}_{j_1, 0}^{0, j_2}(z, J_1, J_2) &:= \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^{j_2}(z, J_1), \text{ for } j_1 \in \{1, 2, \dots, 2J_1\}. \end{aligned}$$

Hence we realize, if we sequentially apply the linear transformation $\tau_{J_1}^{(A)}$ along each column of the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$, we get the matrix-valued statistic:

$$\tilde{\mathbf{c}}_{(1:2J_1), 0}^{0, (1:J_2)}(z, J_1, J_2) = \{ \{ \tilde{\mathbf{c}}_{j_1, 0}^{0, j_2}(z, J_1, J_2) \} \}_{(j_1, j_2) \in \mathbb{N}_{\{J_1, J_2\}}}.$$

Now, let us look at the j_1^{th} row of the matrix $\tilde{\mathbf{c}}_{(1:2J_1), 0}^{0, (1:J_2)}(z, J_1, J_2)$, which is the array:

$$\begin{aligned} \tilde{\mathbf{c}}_{j_1, 0}^{0, (1:J_2)}(z, J_1, J_2) &= (\mathbf{c}_{j_1, 0}^{0, 1}(z, J_1, J_2), \dots, \mathbf{c}_{j_1, 0}^{0, J_2}(z, J_1, J_2)) \\ &= (\mathbf{c}_{j_1}(z) \cdot \tilde{\mathbf{c}}^1(z, J_1), \dots, \mathbf{c}_{j_1}(z) \cdot \tilde{\mathbf{c}}^1(z, J_1)) \\ &= \mathbf{c}_{j_1}(z) \cdot (\tilde{\mathbf{c}}^1(z, J_1), \dots, \tilde{\mathbf{c}}^1(z, J_1)) = \mathbf{c}_{j_1}(z) \cdot \tilde{\mathbf{c}}^{(1:J_2)}(z, J_1). \end{aligned}$$

Again, we realize $\tilde{\mathbf{c}}_{j_1, 0}^{0, (1:J_2)}(z, J_1, J_2)$ is a scalar-multiple of the array $\tilde{\mathbf{c}}^{(1:J_2)}(z, J_1)$ with multiplier $\mathbf{c}_{j_1}(z)$. Now, using the transformation $\tau_J^{(B)}$ in [Lemma 1.2](#) (with $J = J_2$), we conclude:

$$\begin{aligned} \tau_{J_2}^{(B)}(\tilde{\mathbf{c}}^{(1:2J_1), j_2}(z, J_1, J_2)) &= \tau_{J_2}^{(B)}(\mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^{(1:J_2)}(z, J_1)) \\ &= \mathbf{c}_{j_1}(z) \cdot \tau_{J_2}^{(B)}(\dot{\mathbf{c}}^{(1:J_2)}(z, J_1)) = \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{(1:J_2)}(z, J_1) \\ &= \tilde{\mathbf{c}}_{j_1, (1:J_2)}(z, J_1, J_2) = (\tilde{\mathbf{c}}_{j_1, 1}(z, J_1, J_2), \dots, \tilde{\mathbf{c}}_{j_1, J_2}(z, J_1, J_2)); \\ \text{when: } \tilde{\mathbf{c}}_{j_1, j_2}(z, J_1, J_2) &:= \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1), \text{ for } j_2 \in \{1, 2, \dots, J_2\}. \end{aligned}$$

So if we sequentially apply the linear transformation $\tau_{J_2}^{(B)}$ along each row of the matrix $\tilde{\mathbf{c}}_{(1:2J_1), 0}^{0, (1:J_2)}(z, J_1, J_2)$, we'll get with the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$. So we have demonstrated a sequence of linear transformation that transforms the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ into the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$. In other words we established the existence of a linear transformation $\tau_{J_1, J_2}^{(A, B)}$, such that:

$$\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) = \tau_{J_1, J_2}^{(A, B)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)). \quad \square$$

Now, observe that for $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$:

$$\mathbf{c}_{(1:2J_1 J_2)}(z) = (\mathbf{c}_1(z), \dots, \mathbf{c}_{2J_1 J_2}(z, J_1, J_2)), \text{ we have the following result:}$$

LEMMA 1.4. *Given integers $J_1 \in \mathbb{N}^+$ and $J_2 \in \mathbb{N}^+$, there exist a linear-transformation $\tau_{J_1, J_2}^{(C)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1 J_2}$, such that:*

$$\mathbf{c}_{(1:2J_1 J_2)}(z) = \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z)) \text{ for } x \in (0, 1).$$

Proof. First, let us adopt to the following notation for $j_2 \in \{1, \dots, J_2\}$:

$$\mathbf{c}_{2J_1(j_2-1)+(1:2J_2)}(z) = (\mathbf{c}_{2J_1(j_2-1)+1}, \dots, \mathbf{c}_{2J_1(j_2-1)+2J_1}).$$

Note that, if we concatenate: $\mathbf{c}_{2J_1(1-1)+(1:2J_2)}(z), \dots, \mathbf{c}_{2J_1(J_2-1)+(1:2J_2)}(z)$ in a single array, we'll get $\mathbf{c}_{(1:2J_1J_2)}(z)$. Now, we are going to prove our proposed statement: $\mathbf{c}_{2J_1(j_2-1)+(1:2J_2)}(z)$ can be obtained from the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z)$ by linear-transformations for $j_2 \in \{1, 2, \dots, J_2\}$, by principles of mathematical induction. For $j_2 = 1$, note that:

$$\mathbf{c}_{2J_1(1-1)+(1:2J_2)}(z) = (\mathbf{c}_1(z), \dots, \mathbf{c}_{2J_1}(z)) = \mathbf{c}_{(1:2J_1)}(z).$$

Also note that the first column of $\tilde{\mathbf{c}}_{2J_1, J_2}(z)$ is:

$$\begin{aligned} & (\tilde{\mathbf{c}}_{1,1}(z, J_1, J_2), \dots, \tilde{\mathbf{c}}_{2J_1,1}(z, J_1, J_2)) \\ &= (\mathbf{c}_1(z) \cdot \dot{\mathbf{c}}_1(z, J_1), \dots, \mathbf{c}_{2J_1}(z) \cdot \dot{\mathbf{c}}_1(z, J_1)) \\ &= (\mathbf{c}_1(z), \dots, \mathbf{c}_{2J_1}(z)) = \mathbf{c}_{(1:2J_1)}(z). \quad (\text{since } \dot{\mathbf{c}}_1(z, J_1) = 1). \end{aligned}$$

So our proposed statement is trivially true. Now we will assume that for some $j_1 \in \{2, 3, \dots, J_2 - 1\}$ our proposed statement is true, and we will prove it for $j_1 + 1$. By out assumption each element in the array:

$$\mathbf{c}_{2J_1(j_2-1)+(1:2J_2)}(z) = (\mathbf{c}_{2J_1(j_2-1)+1}, \dots, \mathbf{c}_{2J_1(j_2-1)+2J_1}).$$

can be obtained from $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z)$ by linear-transformations. Note that for any $x \in (0, 1)$, $\{J_1, J_2\} \in \mathbb{N}^+$ and $j_1 \in \{1, 2, \dots, J_1\}$, $j_2 \in \{1, 2, \dots, J_2\}$, we have the following trigonometric identities:

$$\begin{aligned} & \cos((2J_1j_2 + (2j_1 - 1))z) + \cos((2J_1j_2 - (2j_1 - 1))z) \\ &= 2\cos(2J_1j_2z)\cos((2j_1 - 1)z), \\ & \sin((2J_1j_2 + (2j_1 - 1))z) - \sin((2J_1j_2 - (2j_1 - 1))z) \\ &= 2\cos(2J_1j_2z)\sin((2j_1 - 1)z). \end{aligned}$$

With little rearrangements, can rewrite these two equations as,

$$\begin{aligned} \mathbf{c}_{2J_1j_2+2j_1-1}(z) &= 2\tilde{\mathbf{c}}_{2j_1-1, j_2}(z, J_1, J_2) - \mathbf{c}_{2J_1j_2-2j_1+1}(z). \\ \mathbf{c}_{2J_1j_2+2j_1}(z) &= 2\tilde{\mathbf{c}}_{2j_1, j_2}(z, J_1, J_2) + \mathbf{c}_{2J_1j_2-2j_1+2}(z). \end{aligned}$$

Observe that, if $j_1 \in \{1, 2, \dots, J_1\}$, then $(2J_1 - 2j_1 + 1) \in \{1, 3, \dots, 2J_1 - 1\}$ and $(2J_1 - 2j_1 + 2) \in \{2, 4, \dots, 2J_1\}$. So, by assumption, both $\mathbf{c}_{2J_1j_2-2j_1+1}(z)$ and $\mathbf{c}_{2J_1j_2-2j_1+2}(z)$ can be obtained from the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_1 \in \{1, 2, \dots, J_1\}$. Then, from the relations mentioned above, we can say that, both $\mathbf{c}_{2J_1j_2+2j_1-1}$ and $\mathbf{c}_{2J_1j_2+2j_1}$ can be obtained from the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_1 \in \{1, 2, \dots, J_1\}$. Hence, the array $\mathbf{c}_{2J_1j_2+(1:2J_1)}(z)$ can be obtained from the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations. So, we proved our induction hypothesis for $j_2 + 1$.

This conclusion proves our induction hypothesis, $\mathbf{c}_{2J_1(j_2-1)+(1:2J_1)}(z)$ can be obtained from $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)$ by linear-transformations for $j_2 \in \{1, 2, \dots, J_2\}$, and hence the concatenated $2J_1J_2$ element array $\mathbf{c}_{(1:2J_1J_2)}(z)$ can be obtained from the matrix $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z)$ by linear transformations. So, we have proved existence of a linear transformation $\tau_J^{(C)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \rightarrow \mathbb{R}^{2J_1J_2}$, such that:

$$\mathbf{c}_{(1:2J_1J_2)}(z) = \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)). \quad \square$$

1.4. K -dimensional generalizations. First we would like to lay out few notations that we are going to use frequently in this section. For a general $K \in \mathbb{N}^+$ and $K > 2$, consider integer vector: $\underline{\mathbf{J}} = (J_1, J_2, \dots, J_K) \in \mathbb{N}^{+K}$. Define integers: $J := \prod_{k=1}^K J_k$ and $L_k := \prod_{l=1}^k J_l$ for $1 \leq k < K$. Also define the index set:

$$\mathbb{N}_{\{\underline{\mathbf{J}}\}} := \{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}.$$

For $\underline{j} = (j_1, j_2, \dots, j_K) \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}$, define K dimensional sequential functions:

$$\tilde{\mathbf{c}}_{\underline{j}}(z, \underline{\mathbf{J}}) = \tilde{\mathbf{c}}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := \mathbf{c}_{j_1}(z) \cdot \prod_{k=2}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}),$$

$$\tilde{\mathbf{c}}^{\underline{j}}(z, \underline{\mathbf{J}}) = \tilde{\mathbf{c}}^{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := \mathbf{c}^{j_1}(z) \cdot \prod_{k=2}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}).$$

We also define K -dimensional array-valued functions:

$$\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{\underline{j}}(z, \underline{\mathbf{J}})\}\}_{\{\underline{j} \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}},$$

$$\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}^{\underline{j}}(z, \underline{\mathbf{J}})\}\}_{\{\underline{j} \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

Then, we have the following result:

LEMMA 1.5. Given $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear transformation

$$\tau_{\underline{\mathbf{J}}}^{(K)} : \mathbb{R}^{2J_1} \times \mathbb{R}^{J_2} \times \dots \times \mathbb{R}^{J_K} \rightarrow \mathbb{R}^{2J} \text{ such that:}$$

$$\mathbf{c}_{(1:2J)}(z) = \tau_{\underline{\mathbf{J}}}^{(K)}(\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Proof. Let us fix the indices $j_2, \dots, j_K$ and traverse through the 1^{st} dimension of the K -dimensional array $\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$. So, we change the first index j_1 over the values: $j_1 = 1, 2, \dots, 2J_1$, while keeping $j_2, \dots, j_K$ fixed and consider every-term $\tilde{\mathbf{c}}^{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ for such combinations of indices in $\mathbb{N}_{\{\underline{\mathbf{J}}\}}$, which is the array:

$$\begin{aligned} & \tilde{\mathbf{c}}^{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{\mathbf{c}}^{1, j_2, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{\mathbf{c}}^{2J_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})) \\ & = \left(\mathbf{c}^1(z) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}), \dots, \mathbf{c}^{2J_1}(z) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}) \right) \\ & = (\mathbf{c}^1(z), \dots, \mathbf{c}^{2J_1}(z)) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}) = \mathbf{c}^{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}). \end{aligned}$$

So, $\tilde{\mathbf{c}}^{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of $\mathbf{c}^{(1:2J_1)}(z)$ with the multiplier $\prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1})$. Then, for $\tau_{J_1}^{(A)}$ in Lemma 1.1 (with $J = J_1$) we get:

$$\begin{aligned} & \tau_{J_1}^{(A)}(\tilde{\mathbf{c}}^{(1:2J_1), j_2, \dots, j_K}(z, \underline{\mathbf{J}})) = \tau_{J_1}^{(A)}\left(\mathbf{c}^{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1})\right) \\ & = \tau_{J_1}^{(A)}(\mathbf{c}^{(1:2J_1)}(z)) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}) = \mathbf{c}_{(1:2J_1)}(z) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_k}(z, L_{k-1}) \\ & = \tilde{\mathbf{c}}_{(1:2J_1), 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{\mathbf{c}}_{1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{\mathbf{c}}_{2J_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}})); \end{aligned}$$

$$\text{where, } \tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}}) := \mathbf{c}_{j_1}(z) \cdot \prod_{k=2}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}),$$

$$\text{for } j_1 \in \{1, 2, \dots, 2J_1\}.$$

Hence, if we sequentially apply $\tau_{J_1}^{(A)}$ along 1st dimension of the K -dimensional array: $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, we'll get the K -dimensional array:

$$\tilde{\mathbf{c}}_{(1:2J_1), 0, \dots, 0}^{0, (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{\{\underline{j} \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

Next, we fix j_1 and $j_3, \dots, j_K$ and traverse through the 2nd dimension of the K -dimensional array: $\mathbf{c}_{(1:2J_1), 0, \dots, 0}^{0, (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, for $j_2 \in \{1, 2, \dots, J_2\}$, which is the array:

$$\begin{aligned} & \tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, 1, \dots, j_K}(z, \underline{\mathbf{J}}), \dots, \tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, J_2, \dots, j_K}(z, \underline{\mathbf{J}})) \\ & = \left(\mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^1(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}), \dots, \right. \\ & \quad \left. \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^{J_2}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}) \right) \\ & = \mathbf{c}_{j_1}(z) \cdot (\dot{\mathbf{c}}^1(z, J_1), \dots, \dot{\mathbf{c}}^{J_2}(z, J_1)) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}) \\ & = \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}). \end{aligned}$$

So, once again, we realize $\tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of the array: $\dot{\mathbf{c}}^{(1:J_2)}(z, J_1)$ with the multiplier $\mathbf{c}_{j_1}(z) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1})$. In other words, this time, for $\tau_{J_2}^{(B)}$ in [Lemma 1.2](#) (with $J = J_2$) we get:

$$\begin{aligned} & \tau_{J_2}^{(B)}(\tilde{\mathbf{c}}_{j_1, 0, \dots, 0}^{0, (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})) \\ & = \tau_{J_2}^{(B)}\left(\mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}^{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1})\right) \\ & = \mathbf{c}_{j_1}(z) \cdot \tau_{J_2}^{(B)}\left(\dot{\mathbf{c}}^{(1:J_2)}(z, J_1)\right) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}) \\ & = \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{(1:J_2)}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_{k-1}) \\ & = \tilde{\mathbf{c}}_{j_1, (1:J_2), \dots, 0}^{0, 0, \dots, j_K}(z, \underline{\mathbf{J}}) = (\tilde{\mathbf{c}}_{j_1, 1, \dots, 0}^{0, 0, \dots, j_p}(z, \underline{\mathbf{J}}), \dots, \tilde{\mathbf{c}}_{j_1, J_2, \dots, 0}^{0, 0, \dots, j_p}(z, \underline{\mathbf{J}})). \end{aligned}$$

$$\text{where, } \tilde{\mathbf{c}}_{j_1, j_2, \dots, 0}^{0, 0, \dots, j_p}(z, \underline{\mathbf{J}}) = \mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}^{j_k}(z, L_k),$$

$$\text{for } j_2 \in \{1, 2, \dots, J_2\}.$$

325 So, if we sequentially apply the linear transformation $\tau_{J_2}^{(B)}$ along the 2^{nd} dimension of
 326 the K -dimensional array: $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}^{(0, 0, \dots, 0)}(z, \underline{\mathbf{J}})$, we will get the K -dimensional
 327 array:

$$328 \quad \tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}^{(0, 0, \dots, 0)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{j_1, j_2, \dots, j_K}^{(0, 0, \dots, 0)}(z, \underline{\mathbf{J}})\}\}_{\{j \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

330 If $K = 3$, and if we similarly apply $\tau_{J_3}^{(B)}$ along the 3^{rd} dimension of the 3-dimensional
 331 array: $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), (1:J_3)}^{(0, 0, 0)}(z, \underline{\mathbf{J}})$, we will get 3-dimensional array:

$$332 \quad \tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), (1:J_3)}^{(0, 0, 0)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{j_1, j_2, j_3}^{(0, 0, 0)}(z, \underline{\mathbf{J}})\}\}_{\{j \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

334 We also realize that, if $K > 3$ and if we recursively keep on applying the linear
 335 transformation $\tau_{J_k}^{(B)}$ along k^{th} dimension of the output arrays for $k = 4, \dots, K$, after
 336 $K - 3$ such transformations, we will end up the the K -dimensional array:

$$337 \quad \tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{\{j \in \mathbb{N}_{\{\underline{\mathbf{J}}\}}\}}.$$

339 Now, let us fix the indices $j_3, \dots, j_K$ and traverse through the 1^{st} and 2^{nd} dimensions
 340 of this K -dimensional array: $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$. We simultaneously change
 341 the first index j_1 over the values: $j_1 = 1, 2, \dots, 2J_1$ along with the second index j_2
 342 over: $j_2 = 1, 2, \dots, J_2$, while we keep $j_3, \dots, j_K$ fixed and we look for every term
 343 $\tilde{\mathbf{c}}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})$ for such combinations of indices in $\mathbb{N}_{\{\underline{\mathbf{J}}\}}$, thus, we get the matrix:

$$\begin{aligned} 344 \quad & \tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}_{j_1, j_2, \dots, j_K}(z, \underline{\mathbf{J}})\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \\ 345 \quad & = \{\{\mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1})\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \\ 346 \quad & = \{\{\mathbf{c}_{j_1}(z) \cdot \dot{\mathbf{c}}_{j_2}(z, J_1)\}\}_{1 \leq j_1 \leq 2J_1, 1 \leq j_2 \leq J_2} \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}) \\ 347 \quad & = \tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}). \end{aligned}$$

349 Then, we realize that $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})$ is a scalar-multiple of the matrix:

350 $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1)$ with multiplier $\prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1})$. Then, for $\tau_{J_1, J_2}^{(C)}$ as in [Lemma 1.4](#)

we get:

$$\begin{aligned}
& \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, j_K}(z, \underline{\mathbf{J}})) \\
&= \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1})) \\
&= \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2)}(z, J_1, J_2)) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}) \\
&= \mathbf{c}_{(1:2J_1 J_2)}(z) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}) \\
&= \tilde{\mathbf{c}}_{(1:2J_1 J_2), j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}) \\
&= (\tilde{\mathbf{c}}_{1, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}), \dots, \tilde{\mathbf{c}}_{2J_1 J_2, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]})); \\
&\text{where } \underline{\mathbf{J}}_{[K-1]} = (J_1 J_2, J_3, \dots, J_K) = (L_2, J_3, \dots, J_K) \in \mathbb{N}^{+K-1}; \text{ and} \\
&\tilde{\mathbf{c}}_{j, j_3, \dots, j_K}(z, \underline{\mathbf{J}}_{[K-1]}) = \mathbf{c}_j(z) \cdot \prod_{k=3}^K \dot{\mathbf{c}}_{j_k}(z, L_{k-1}) \\
&\hspace{15em} \text{for } j \in \{1, \dots, 2J_1 J_2\}.
\end{aligned}$$

From our notation, we have:

$$\mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}} = \{1, \dots, 2J_1 J_2\} \times \dots \times \{1, \dots, J_K\} \subset \mathbb{N}^{+K-1}$$

A general $\mathbf{j} = (j, j_3, \dots, j_K) \in \mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}}$ is a $K - 1$ dimensional integer vector. Hence, if we apply the linear transformation $\tau_{J_1, J_2}^{(C)}$ along 1^{st} and 2^{nd} dimensions of $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$, while keeping the other indices fixed we will get the $K - 1$ -dimensional array:

$$\tilde{\mathbf{c}}_{(1:2J_1 J_2), (1:J_3), \dots, (1:J_K)}(z, \underline{\mathbf{J}}_{[K-1]}) = \{\{\tilde{\mathbf{c}}_j(z, \underline{\mathbf{J}}_{[K-1]})\}\}_{\{j \in \mathbb{N}_{\{\underline{\mathbf{J}}_{[K-1]}\}}\}}.$$

If $K = 3$, at this stage we have the matrix valued statistic $\tilde{\mathbf{c}}_{(1:2J_1 J_2), (1:J_3)}(z, \underline{\mathbf{J}}_{[2]})$, where $\underline{\mathbf{J}}_{[2]} = (J_1 J_2, J_3)$. If we similarly apply the linear transformation $\tau_{J_1 J_2, J_3}^{(C)}$ along 1^{st} and 2^{nd} dimensions of $\tilde{\mathbf{c}}_{(1:2J_1 J_2), (1:J_3)}(z, \underline{\mathbf{J}}_{[2]})$, we will get the array of length $J = \prod_{k=1}^3 J_k$, which is: $\mathbf{c}_{(1:2J)}(z)$. We also realize, for $K > 3$ if we recursively keep on applying the linear transformation $\tau_{L_k, J_{k+1}}^{(C)}$ along 1^{st} and 2^{nd} dimensions of the output arrays, after $K - 3$ such transformations, we will finally end up with the array of length $J = \prod_{k=1}^K J_k$, which is: $\mathbf{c}_{(1:2J)}(z)$. So we have demonstrated a sequence of linear transformation that transforms the K -dimensional array $\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})$ into the array $\mathbf{c}_{(1:2J)}(z)$. In other words we have established the existence of a linear transformation $\tau_{\underline{\mathbf{J}}}^{(K)}$ so that:

$$\mathbf{c}_{(1:2J)}(z) = \tau_{\underline{\mathbf{J}}}^{(K)}(\tilde{\mathbf{c}}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})). \quad \square$$

Remember, we defined $c_0(z) = 1$, let us define:

$$\mathbf{c}_{(0:2J)}(z) = (c_0(z), c_1(z), \dots, c_{2J}(z)) = (1, \mathbf{c}_{(1:2J)}(z)).$$

$\mathbf{c}_{(0:2J)}(z)$ is indexed by the index set $\mathbb{N}_J = \{0, 1, \dots, 2J\}$. Now remember, for $\underline{\mathbf{J}} = (J_1, \dots, J_K)$, we defined the index set:

$$\mathbb{N}_{\underline{\mathbf{J}}} = \{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}.$$

We add the element 0 to this index set to define a new index set:

$$\mathbb{N}_{\underline{\mathbf{J}}} = \{0\} \cup (\{1, \dots, 2J_1\} \times \{1, \dots, J_2\} \times \dots \times \{1, \dots, J_K\}).$$

Also, for the new index element 0, define: $\tilde{\mathbf{c}}^0(z, \underline{\mathbf{J}}) = 1$. Now we define:

$$\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}}) = \{\{\tilde{\mathbf{c}}^{\underline{\mathbf{j}}}(z, \underline{\mathbf{J}})\}\}_{\{\underline{\mathbf{j}} \in \mathbb{N}_{\underline{\mathbf{J}}}\}}$$

Then, we have the following lemma:

LEMMA 1.6. For $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation

$$\mathcal{T}_{\underline{\mathbf{J}}}^{(K)} : \mathbb{R}^{2J_1} \times \dots \times \mathbb{R}^{J_K} \rightarrow \mathbb{R}^{2J} \text{ such that:}$$

$$\mathbf{c}_{(0:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Proof. If $K = 1$ then $\underline{\mathbf{J}} = (J_1)$ and $J = J_1$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{J_1}^{(A)}(\mathbf{v})$, here as $\tau_{J_1}^{(A)}$ is same as in Lemma 1.1 (for $J = J_1$). Then we have:

$$\mathbf{c}_{(1:2J)}(z) = \mathbf{c}_{(1:2J_1)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{c}_{(1:2J_1)}(z)) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\tilde{\mathbf{c}}^{(1:2J_1)}(z, \underline{\mathbf{J}})).$$

If $K = 2$ then $\underline{\mathbf{J}} = (J_1, J_2)$ and $J = J_1 J_2$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{J_1, J_2}^{(C)}(\tau_{J_1, J_2}^{(A, B)}(\mathbf{v}))$, here $\tau_{J_1, J_2}^{(A, B)}$ is same as in Lemma 1.3 and $\tau_{J_1, J_2}^{(C)}$ is same as in Lemma 1.4. Then we have:

$$\begin{aligned} \mathbf{c}_{(1:2J)}(z) &= \tau_{J_1, J_2}^{(C)}(\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2)) \\ &= \tau_{J_1, J_2}^{(C)}\left(\tau_{J_1, J_2}^{(A, B)}(\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, J_1, J_2))\right) \\ &\Leftrightarrow \mathbf{c}_{(1:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\tilde{\mathbf{c}}^{(1:2J_1), (1:J_2)}(z, \underline{\mathbf{J}})). \end{aligned}$$

If $K > 2$ then $\underline{\mathbf{J}} = (J_1, J_2, \dots, J_K)$ and $J = \prod_{k=1}^K J_k$. Take $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{v}) = \tau_{\underline{\mathbf{J}}}^{(K)}(\mathbf{v})$, here $\tau_{\underline{\mathbf{J}}}^{(K)}$ is same as in Lemma 1.5

$$\mathbf{c}_{(1:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{c}_{(1:2J_1), (1:J_2), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Then we have the generic linear-transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'} : \mathbb{R}^{2J_1} \times \dots \times \mathbb{R}^{J_K} \rightarrow \mathbb{R}^{2J}$ for $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$ satisfying:

$$\mathbf{c}_{(1:2J)}(z) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}(\mathbf{c}_{(1:2J_1), \dots, (1:J_K)}(z, \underline{\mathbf{J}})).$$

Now, observe that for $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, we can break down the generic transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K)'}$, into small simple components: $\tau_J^{(A)}$, $\tau_J^{(B)}$, $\tau_{J_1, J_2}^{(A, B)}$ and $\tau_{J_1, J_2}^{(C)}$. All of these simple transformation gives a linear-combination of the input vector as the

output. Suppose $\mathcal{V}$ is the generic invertible transformation that vectorizes a multi-dimensional array in the natural-order of the index-set of the array. Then, we will be able to find a $2J \times 2J$ matrix $\mathcal{A}_{\underline{J}}^{(K) \prime}$, such that:

$$\mathbf{c}_{(1:2J)}(z) = \mathcal{A}_{\underline{J}}^{(K) \prime} \cdot \mathcal{V}(\mathbf{c}^{(1:2J_1), \dots, (1:J_K)}(z, \mathbf{J})).$$

Now observe that:

$$\mathcal{V}(\mathbf{c}^{0; (1:2J_1), \dots, (1:J_K)}(z, \mathbf{J})) = \left(1, \mathcal{V}(\mathbf{c}^{(1:2J_1), \dots, (1:J_K)}(z, \mathbf{J}))\right)$$

Take $\mathcal{A}_{\underline{J}}^{(K)} := \text{diag}(1, \mathcal{A}_{\underline{J}}^{(K) \prime})$, since $\mathbf{c}_{(0:2J)}(z) = (1, \mathbf{c}_{(1:2J)}(z))$, we must have

$$\mathbf{c}_{(0:2J)}(z) = \mathcal{A}_{\underline{J}}^{(K)} \cdot \mathcal{V}(\mathbf{c}^{0; (1:2J_1), \dots, (1:J_K)}(z, \mathbf{J})).$$

So, we have proved the existence of a transformation $\mathcal{T}_{\underline{J}}^{(K)}$, such that:

$$\mathbf{c}_{(0:2J)}(z) = \mathcal{T}_{\underline{J}}^{(K)}(\mathbf{c}^{0; (1:2J_1), \dots, (1:J_K)}(z, \mathbf{J})). \quad \square$$

Now consider a p -dimensional vector $\mathbf{x} \in (\mathbf{0}, \mathbf{1})$, so that if $\mathbf{x} = (x_1, \dots, x_p)$, then $x_l \in (0, 1)$ for $1 \leq l \leq p$. Let us define the p -dimensional function:

$$\mathbf{c}_{(0:2J); p}(\mathbf{x}) = \otimes_{l=1}^p \mathbf{c}_{(0:2J)}(x_l).$$

Note that $\mathbf{c}_{(0:2J); p}(\mathbf{x})$ can be identified as a p -dimensional array with dimensions $(2J+1) \times \dots \times (2J+1)$ (p times). This p -dimensional array is indexed by the index set:

$$\mathbb{N}_{J; p} = \{0, \dots, 2J\} \times \dots \times \{0, \dots, 2J\} \text{ (} p \text{ times)}.$$

For $\mathbf{j} = (j_1, \dots, j_p) \in \mathbb{N}_{J; p}$, a general $\mathbf{j}^{th}$ element of $\mathbf{c}_{(0:2J); p}(\mathbf{x})$ is the function:

$$\mathbf{c}_{\mathbf{j}}(\mathbf{x}) = \prod_{l=1}^p \mathbf{c}_{j_l}(x_l).$$

Let us also define:

$$\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(\mathbf{x}, \underline{\mathbf{J}}) = \otimes_{l=1}^p \left(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{\mathbf{J}}) \right).$$

This array-like function is indexed by the index set: $\mathbb{N}_{\underline{\mathbf{J}}; p}$, defined as:

$$\mathbb{N}_{\underline{\mathbf{J}}; p} = \mathbb{N}_{\underline{\mathbf{J}}} \times \dots \times \mathbb{N}_{\underline{\mathbf{J}}} \text{ (} p \text{ times)}.$$

Let us denote a general index element of $\mathbb{N}_{\{\underline{\mathbf{J}}, p\}}$ as $\underline{\mathbf{j}}$, which is a p -tuple: $(\underline{\mathbf{j}}_1, \dots, \underline{\mathbf{j}}_p)$.

For $1 \leq l \leq p$, $\underline{\mathbf{j}}_l$ is either 0 or a K -element-vector $(j_{l,1}, \dots, j_{l,K})$

Now, the $\underline{\mathbf{j}}^{th}$ element of $\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(\mathbf{x}, \underline{\mathbf{J}})$ is:

$$\tilde{\mathbf{c}}^{\underline{\mathbf{j}}}(\mathbf{x}, \underline{\mathbf{J}}) = \prod_{l=1}^p \left(\mathbf{c}^{j_{l,1}}(x_l) \cdot \prod_{k=2}^K \tilde{\mathbf{c}}^{j_{l,k}}(x_l, L_{k-1}) \right)^{1(\underline{\mathbf{j}}_l \neq 0)}.$$

We have the following lemma:

LEMMA 1.7. Let $\mathbf{x}$ is an element of the p -dimensional open interval $(\mathbf{0}, \mathbf{1})$, for $K \in \mathbb{N}^+$ and $\underline{J} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation $\mathcal{T}_{\underline{J}}^{(K,p)}$, such that:

$$\mathbf{c}_{(0:2J)^p}(\mathbf{x}) = \mathcal{T}_{\underline{J}}^{(K,p)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}, \underline{J})).$$

Proof. From Lemma 1.6, we have for $1 \leq l \leq p$:

$$\mathbf{c}_{(0:2J)}(x_l) = \mathcal{T}_{\underline{J}}^{(K)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{J})).$$

If we take tensor-product of both sides for $1 \leq l \leq p$, we get:

$$\begin{aligned} \otimes_{l=1}^p \mathbf{c}_{(0:2J)}(x_l) &= \otimes_{l=1}^p \left(\mathcal{T}_{\underline{J}}^{(K)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{J})) \right) \\ \Leftrightarrow \mathbf{c}_{(0:2J)^p}(\mathbf{x}) &= \left(\otimes_{l=1}^p \mathcal{T}_{\underline{J}}^{(K)} \right) \left(\otimes_{l=1}^p (\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)}(x_l, \underline{J})) \right) \\ \Leftrightarrow \mathbf{c}_{(0:2J)^p}(\mathbf{x}) &= \mathcal{T}_{\underline{J}}^{(K,p)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}, \underline{J})). \end{aligned}$$

If we let $\mathcal{T}_{\underline{J}}^{(K,p)} := \otimes_{l=1}^p \mathcal{T}_{\underline{J}}^{(K)}$, we have:

$$\mathbf{c}_{(0:2J)^p}(\mathbf{x}) = \mathcal{T}_{\underline{J}}^{(K,p)}(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}, \underline{J})). \quad \square$$

1.5. Main theorem. Now remember, we have the following SEP statistic and standardized WEP statistic for $\underline{j} \in \mathbb{N}_{J;p}$:

$$\mathbf{C}_j(\mathbf{X}) = \sum_{i \in I} \mathbf{c}_j(\mathbf{x}_i), \quad \bar{\mathbf{C}}_j(\mathbf{X}) = (1/\mathbf{x}) \cdot \mathbf{C}_j(\mathbf{X}).$$

Let us define the collections:

$$\begin{aligned} \mathbf{C}_{(0:2J)^p}(\mathbf{X}) &= \{ \{ \mathbf{C}_j(\mathbf{X}) \} \}_{j \in \mathbb{N}_{J;p}} \\ \bar{\mathbf{C}}_{(0:2J)^p}(\mathbf{X}) &= \{ \{ \bar{\mathbf{C}}_j(\mathbf{X}) \} \}_{j \in \mathbb{N}_{J;p}}. \end{aligned}$$

Note that:

$$\mathbf{C}_{(0:2J)^p}(\mathbf{X}) = \sum_{i \in I} \mathbf{c}_{(0:2J)^p}(\mathbf{x}_i), \quad \bar{\mathbf{C}}_{(0:2J)^p}(\mathbf{X}) = (1/\mathbf{x}) \cdot \mathbf{C}_{(0:2J)^p}(\mathbf{X}).$$

We define the following SEP statistic and standardized WEP statistic for a general $\underline{j} \in \mathbb{N}_{\underline{J};p}$:

$$\tilde{\mathbf{C}}^{\underline{j}}(\mathbf{X}, \underline{J}) = \sum_{i \in I} \tilde{\mathbf{c}}^{\underline{j}}(\mathbf{x}_i, \underline{J}), \quad \bar{\tilde{\mathbf{C}}}^{\underline{j}}(\mathbf{X}, \underline{J}) = (1/\mathbf{x}) \cdot \tilde{\mathbf{C}}^{\underline{j}}(\mathbf{X}, \underline{J}).$$

Again, we define the collections of statistics:

$$\begin{aligned} \tilde{\mathbf{C}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{J}) &= \{ \{ \tilde{\mathbf{C}}^{\underline{j}}(\mathbf{X}, \underline{J}) \} \}_{\underline{j} \in \mathbb{N}_{\underline{J};p}} \\ \bar{\tilde{\mathbf{C}}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{J}) &= \{ \{ \bar{\tilde{\mathbf{C}}}^{\underline{j}}(\mathbf{X}, \underline{J}) \} \}_{\underline{j} \in \mathbb{N}_{\underline{J};p}}. \end{aligned}$$

And, we have the identities:

$$\tilde{\mathbf{C}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{J}) = \sum_{i \in I} \tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}_i, \underline{J})$$

$$\bar{\tilde{\mathbf{C}}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{J}) = (1/\mathbf{x}) \cdot \tilde{\mathbf{C}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{J}).$$

Now, we are in a position to prove the main-result of this supplementary text:

503 THEOREM 1.8. For $K \in \mathbb{N}^+$ and $\underline{\mathbf{J}} \in \mathbb{N}^{+K}$, $\exists$ a linear-transformation $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}$,
 504 such that:

$$505 \quad \bar{\mathbf{C}}_{(0:2J)^p}(\mathbf{X}) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}\left(\bar{\tilde{\mathbf{C}}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{\mathbf{J}})\right).$$

507 *Proof.* From Lemma 1.7, we have for $i \in I$:

$$508 \quad \mathbf{c}_{(0:2J)^p}(\mathbf{x}_i) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}\left(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}_i, \underline{\mathbf{J}})\right).$$

510 Since $\mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}$ is a linear-transformation, if we take a sum over $i \in I$ and divide both
 511 sides by $|\mathbf{X}|$, we get:

$$\begin{aligned} 512 \quad & \frac{\sum_{i \in I} \mathbf{c}_{(0:2J)^p}(\mathbf{x}_i)}{|\mathbf{X}|} = \frac{\sum_{i \in I} \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}\left(\tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}_i, \underline{\mathbf{J}})\right)}{|\mathbf{X}|} \\ 513 \quad & \Leftrightarrow \frac{\sum_{i \in I} \mathbf{c}_{(0:2J)^p}(\mathbf{x}_i)}{|\mathbf{X}|} = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}\left(\frac{\sum_{i \in I} \tilde{\mathbf{c}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{x}_i, \underline{\mathbf{J}})}{|\mathbf{X}|}\right) \\ 514 \quad & \Leftrightarrow \bar{\mathbf{C}}_{(0:2J)^p}(\mathbf{X}) = \mathcal{T}_{\underline{\mathbf{J}}}^{(K,p)}\left(\bar{\tilde{\mathbf{C}}}^{0; (1:2J_1), (1:J_2), \dots, (1:J_K)^p}(\mathbf{X}, \underline{\mathbf{J}})\right). \quad \square \\ 515 \end{aligned}$$